

LoRaWAN Using RAK3172 and TTN

1 RAK3172

1.1 Hardware

- Pin 1: UART2_RX (input) AT Commands and FW Update
- Pin 2: UART2_TX (output) AT Commands and FW Update
- Pin 24: 3.3V
- Pin 28: GND

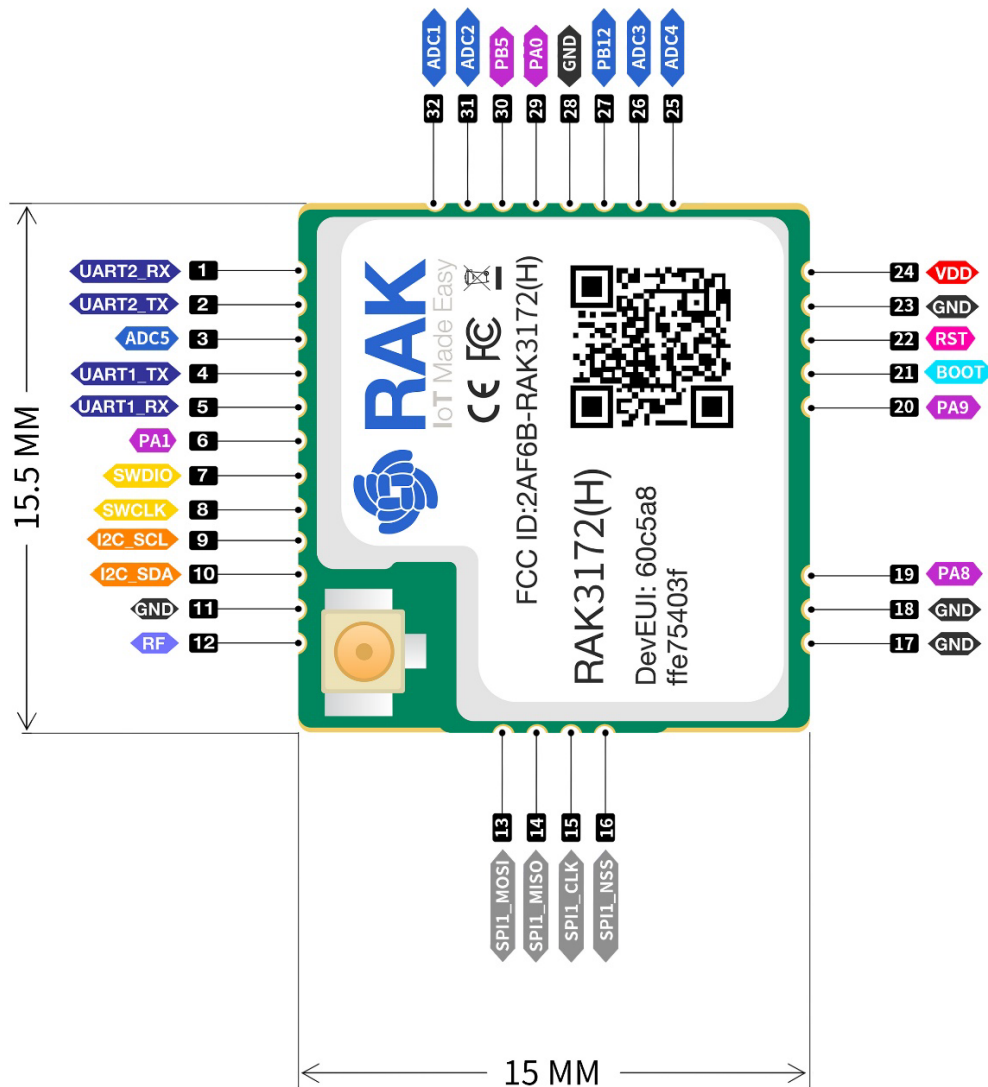


Figure 1: RAK3172 Pinout

1.2 Setup/Configuration

From the OmniWAN console:

@wan send ATR	Restore factory defaults
AT+NWM=1	Network work mode must be 1 for LoRaWAN
AT+NJM=1	Join Mode: 0:ABP, 1:OTAA
AT+CLASS=?	Class A for lowest power
AT+BAND=5	US915 band
	Data Rate
AT+DEVEUI=?	DevEUI (8 hex bytes) (for OTAA)
AT+APPEUI=?	AppEUI (8 hex bytes) (for OTAA)
AT+APPKEY=?	AppKEY (16 hex bytes) (for OTAA)
AT+DEVADDR=?	DevAddr (4 hex bytes) (for ABP)
AT+APPSKEY=?	AppSKey (16 hex bytes) (for ABP)
AT+NWKSKEY=?	NwkSKey (16 hex bytes) (for ABP)

LoRaWAN version ⓘ *

Regional Parameters version ⓘ *

Frequency plan ⓘ *

[Show advanced activation, LoRaWAN class and cluster settings](#) ▼

DevEUI ⓘ *

 0/50 used

AppEUI ⓘ *

AppKey ⓘ *

End device ID ⓘ *

This value is automatically prefilled using the DevEUI

1.3 Frequency Bands and the Mask

For FSB2 (TTN default channels): Set the mask as:

AT+MASK=0002

US915 uplink Channels									
902.3	902.5	902.7	902.9	903.1	903.3	903.5	903.7	Channels0 to 7	125kHz DR0 to DR3 4/5
903.9	904.1	904.3	904.5	904.7	904.9	905.1	905.3	Channels8 to 15	
905.5	905.7	905.9	906.1	906.3	906.5	906.7	906.9	Channels16 to 23	
907.1	907.3	907.5	907.7	907.9	908.1	908.3	908.5	Channels24 to 31	
908.7	908.9	909.1	909.3	909.5	909.7	909.9	910.1	Channels32 to 39	
910.3	910.5	910.7	910.9	911.1	911.3	911.5	911.7	Channels40 to 47	
911.9	912.1	912.3	912.5	912.7	912.9	913.1	913.3	Channels48 to 55	
913.5	913.7	913.9	914.1	914.3	914.5	914.7	914.9	Channels56 to 63	
903	904.6	906.2	907.8	909.4	911	912.6	914.2	Channels64 to 71	500kHz DR4

Gateway Frequencies:

Gateway Channels frequency

chan_multSF_0
Lora MAC, 125kHz, all SF, 903.9 MHz

chan_multSF_1
Lora MAC, 125kHz, all SF, 904.1 MHz

chan_multSF_2
Lora MAC, 125kHz, all SF, 904.3 MHz

chan_multSF_3
Lora MAC, 125kHz, all SF, 904.5 MHz

chan_multSF_4
Lora MAC, 125kHz, all SF, 904.7 MHz

chan_multSF_5
Lora MAC, 125kHz, all SF, 904.9 MHz

chan_multSF_6
Lora MAC, 125kHz, all SF, 905.1 MHz

chan_multSF_7
Lora MAC, 125kHz, all SF, 905.3 MHz

chan_Lora_std
Lora MAC, 500kHz, SF8, 904.6 MHz

1.4 Sending to TTN

Summary:

AT+DEVEUI=ac1f09fffe05325d

AT+APPEUI=ac1f09fff8683172

AT+APPKEY=ac1f09fffe05325dac1f09fff8683172

AT+MASK=0002

AT+SN=?

6310C0205010501552

OK

AT+DEVEUI=?

ac1f09fffe05325d

OK

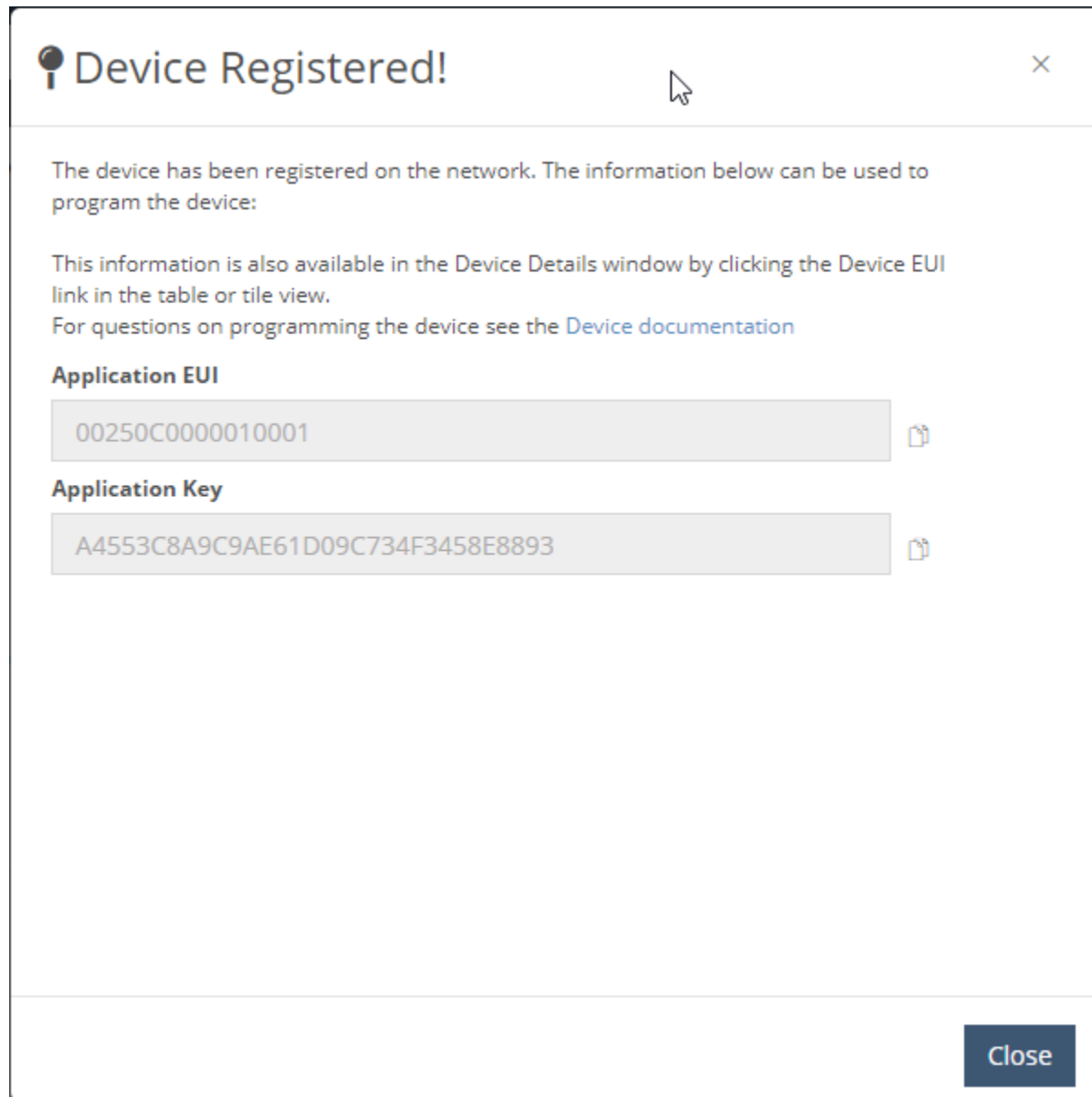
AT+APPEUI=?

00250c0000010001

OK

AT+APPKEY=?

a4553c8a9c9ae61d09c734f3458e8893



Senet Developer API Key: AK3:UQ3SligPV9DFBGazRsQGGB5KfYxcGyqcjGeDvK6m

Use the RAK serial tool:

Command			
☒	01	AT	SEND
☒	02	AT+SN=?	SEND
☒	03	AT+DEVEUI=?	SEND
☒	04	AT+APPEUI=?	SEND
☒	05	AT+APPKEY=?	SEND
☒	06	AT+NJM=?	SEND
☒	07	AT+CFM=?	SEND
☒	08	AT+BAND=?	SEND
☒	09	AT+VER=?	SEND
☒	10	AT+NWM=1	SEND
☒	11	AT+NJM=1	SEND
☒	12	AT+CLASS=A	SEND
☒	13	AT+BAND=5	SEND
☒	14	AT+MASK=0002	SEND
☒	15	AT+JOIN=1:0:10:8	SEND
☒	16	AT+SEND=2:FACEFEED	SEND
☒	17		SEND
☒	18	AT+TXP=?	SEND
☒	19	at+get_config=lora:status	SEND
☒	20	at+get_config=lora:channel	SEND

2 Dragino LIG16 LoRaWAN Gateway Setup

Must connect via WiFi to configure:

 dragino-1ea5bc
Connected, secured

[Properties](#)

Disconnect

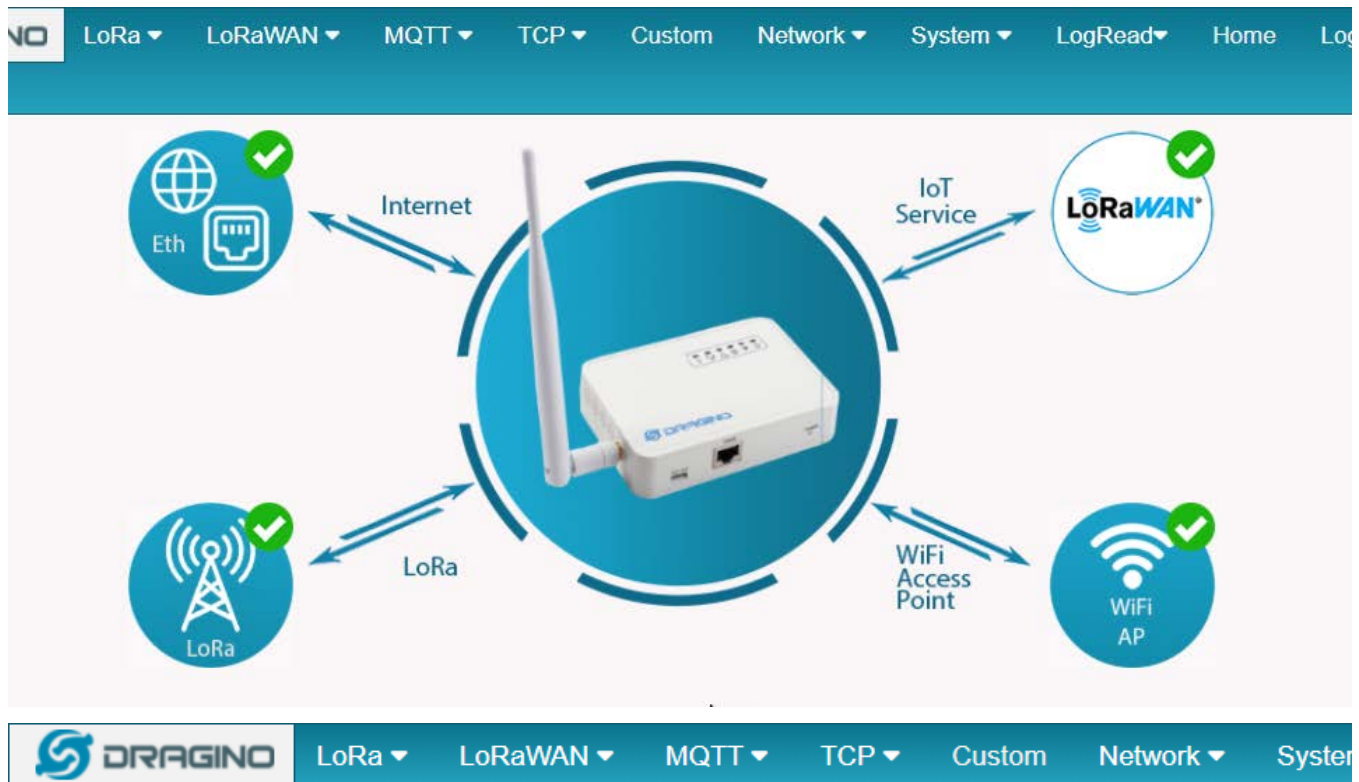
UI at: 10.130.1.1

User is "root", no password set

Defaults:

Email: dragino-1ea5bc@dragino.com

Gateway ID a84041ffff1ea5bc (default)



LoRa Configuration

Debug Level

Radio Settings

Keep Alive Period (sec)
Frequency Plan
Frequency Sub Band

Static GPS coordinates ?

Enable Static GPS ☒
Latitude
Altitude (m)
Longitude

Current Mode: **LoRaWAN Semtech UDP**

LoRaWAN Configuration

General Settings

Email

Gateway ID

Primary LoRaWAN Server

Service Provider Server Address

Uplink Port Downlink Port

Packet Filter

Fport Filter ? DevAddr Filter ?

Current Mode: **LoRaWAN Semtech UDP**

Network

LAN Settings

IP Address Gateway

Netmask DNS

WAN Settings

Enable DHCP

WiFi WAN Settings

Enable DHCP

WiFi

Radio Settings

Channel (1-11)

11

Tx Power (0-18) dBm

17

WiFi Access Point Settings

Enable WiFi Access Point ☒

WiFi Name SSID

dragino-1ea5bc

Passphrase (8-32 char)

dragino+dragino

Hide

Encryption

WPA2

WiFi WAN Client Settings

Enable WiFi WAN Client ☐

Host WiFi SSID

Host-SSID

WiFi Survey

Choose WIFI SSID...

Passphrase

Host-Passphrase

Hide

Encryption

WPA2

Save&Apply

Cancel

Refresh

System General

System Password

Password

Hide

SetPassword

Login: root

Password (admin)

Hide

SetAdminPassword

TimeZone

Timezone

(GMT -5:00) Eastern Time (US & Canada), Bogota, Lima

Port Forwarding

Enable HTTP Forward ☒

Enable SSH Forward ☒

Save&Apply

Cancel





LoRa ▼
LoRaWAN ▼
MQTT ▼
TCP ▼
Custom

System Overview

Device Model: LIG16

Hostname: dragino-1ea5bc

Firmware: lgw-5.4.1622514596

Build Time: Build Tue Jun 1 10:29:56 CST 2021

FWD version: Release:2021-04-07 10:40:58, Version:2.0.6

Cellular : Not Detected

System Time: Mon Dec 13 21:25:49 EST 2021

Uptime: 140 days

Load Avg: 56, load average

Memory: Free Memory: 19904 / Total Memory: 60192kB

IoT Service: lorawan

ETH0 MAC: A8:40:41:1E:A5:BF

ETH1 MAC: A8:40:41:1E:A5:BE

WiFi MAC: A8:40:41:1E:A5:BC

Internet Connection OK



LoRaWAN Connection OK



The actual frequencies are listed at:

<http://10.130.1.1/cgi-bin/log-lora.has>

```

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-----
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chan_multSF_4
Lora MAC, 125kHz, all SF, 904.7 MHz
-----
chan_multSF_5
Lora MAC, 125kHz, all SF, 904.9 MHz

```

```
-----  
chan_multSF_6  
Lora MAC, 125kHz, all SF, 905.1 MHz  
-----  
chan_multSF_7  
Lora MAC, 125kHz, all SF, 905.3 MHz  
-----  
chan_Lora_std  
Lora MAC, 500kHz, SF8, 904.6 MHz
```